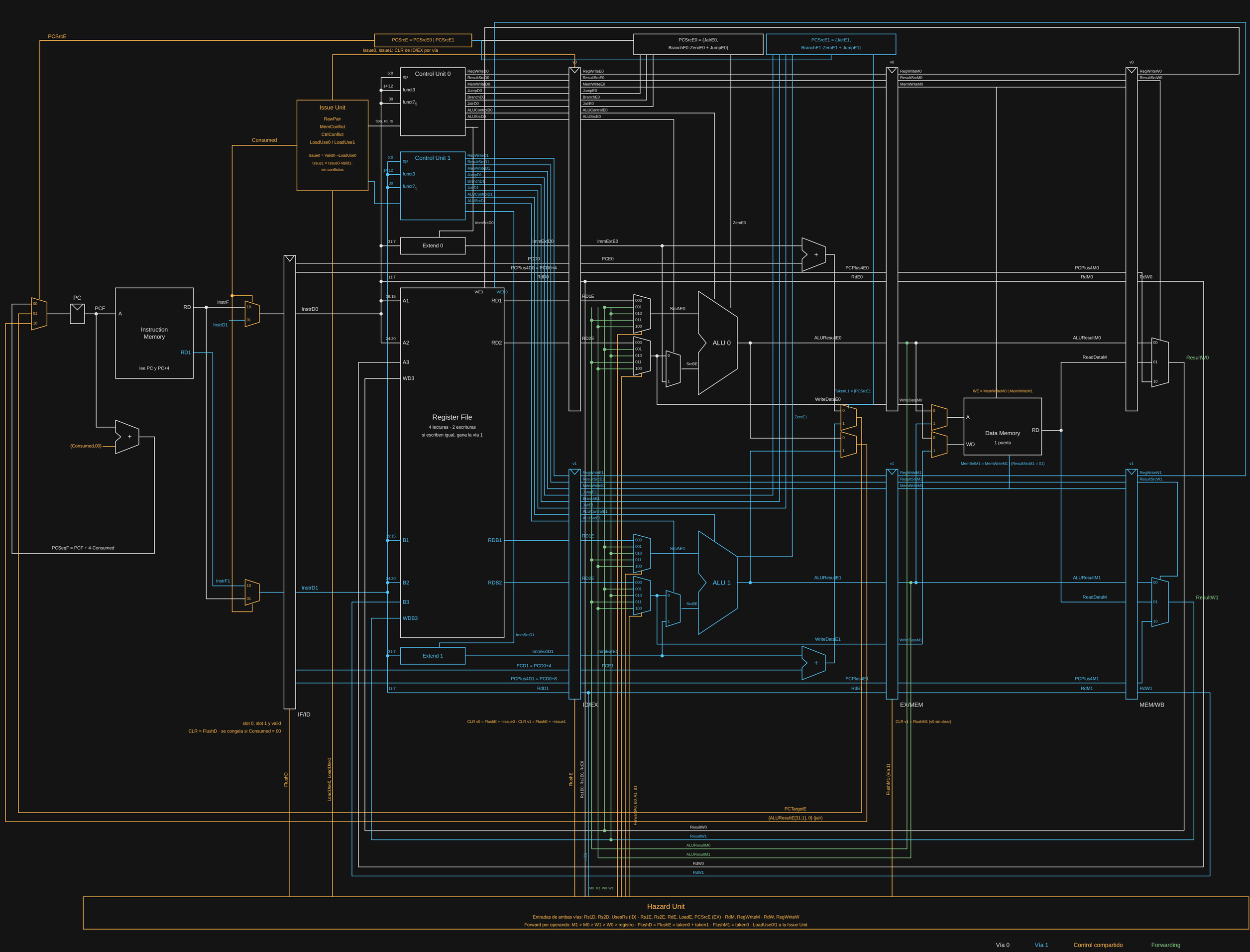

PCSrcE
PCSrcE = PCSrcE0 | PCSrcE1
Issue0, Issue1: CLR de ID/EX por vía
PCSrcE0 = {JalrE0, BranchE0·ZeroE0 + JumpE0}
PCSrcE1 = {JalrE1, BranchE1·ZeroE1 + JumpE1}
Issue Unit
RawPair
MemConflict
CtrlConflict
LoadUse0 / LoadUse1
Issue0 = Valid0·¬LoadUse0
Issue1 = Issue0·Valid1·
sin conflictos
Consumed
Control Unit 0
Control Unit 1
op
funct3
funct7₅
6:0
14:12
30
rd0, rd1, rs
RegWrite0
ResultSrc0
MemWrite0
Jump0
Branch0
Jalr0
ALUControl0
ALUSrc0
RegWrite1
ResultSrc1
MemWrite1
Jump1
Branch1
Jalr1
ALUControl1
ALUSrc1
ImmSrc0
ImmSrc1
Extend 0
Extend 1
31:7
ImmExt0
ImmExt1
PCD0
PCD1
PCPlus4D0 = PCD0+4
PCPlus4D1 = PCD0+8
RdD0
RdD1
11:7
WE3
WE3
19:15
24:20
A1
A2
A3
WD3
RD1
RD2
Register File
4 lecturas · 2 escrituras
si escriben igual, gana la vía 1
PC
PCF
A
RD
Instruction Memory
lee PC y PC+4
InstrF
InstrD0
InstrF1
InstrD1
00
01
10
{Consumed,00}
PCSeqF = PCF + 4·Consumed
v0
v1
RegWriteE0
ResultSrcE0
MemWriteE0
JumpE0
BranchE0
JalrE0
ALUControlE0
ALUSrcE0
RegWriteE1
ResultSrcE1
MemWriteE1
JumpE1
BranchE1
JalrE1
ALUControlE1
ALUSrcE1
ZeroE0
ZeroE1
RD1E
RD2E
SrcAE0
SrcAE1
SrcBE
ALU 0
ALU 1
000
001
010
011
100
ImmExtE0
ImmExtE1
PCE0
PCE1
PCPlus4E0
PCPlus4E1
RdE0
RdE1
ALUResultE0
ALUResultE1
WriteDataE0
WriteDataE1
TakenL1 = |PCSrcE1
RegWriteM0
ResultSrcM0
MemWriteM0
RegWriteM1
ResultSrcM1
MemWriteM1
ALUResultM0
ALUResultM1
WriteDataM0
WriteDataM1
PCPlus4M0
PCPlus4M1
RdM0
RdM1
WE = MemWriteM0 | MemWriteM1
Data Memory
1 puerto
WD
ReadDataM
MemSelM1 = MemWriteM1 · (ResultSrcM1 = 01)
RegWriteW0
ResultSrcW0
RdW0
ResultW0
RegWriteW1
ResultSrcW1
RdW1
ResultW1
IF/ID
ID/EX
EX/MEM
MEM/WB
slot 0, slot 1 y valid
CLR = FlushD · se congela si Consumed = 00
CLR v0 = FlushE + ~Issue0 · CLR v1 = FlushE + ~Issue1
CLR v1 = FlushM1 (v0 sin clear)
FlushD
LoadUse0\ LoadUse1
FlushE
Rs1E0, Rs2E0, RdE0
ForwardA0, B0, A1, B1
FlushM1 (vía 1)
PCTargetE
{ALUResultE[31:1], 0} (jalr)
ResultW0
ResultW1
ALUResultM0
ALUResultM1
RdW0
RdW1
M0 M1 W0 W1
Hazard Unit
Entradas de ambas vías: Rs1D, Rs2D, UsesRs (ID) · Rs1E, Rs2E, RdE, LoadE, PCSrcE (EX) · RdM, RegWriteM · RdW, RegWriteW
Forward por operando: M1 > M0 > W1 > W0 > registro · FlushD = FlushE = taken0 + taken1 · FlushM1 = taken0 · LoadUse0/1 a la Issue Unit
Vía 0
Vía 1
Control compartido
Forwarding